Acceptance Test Plan

Operational model 01_Operational_Merged and the strategic models, Camunda 8 (self-managed, c8run), executed 2026-09-26T16:17:23.628Z.

Result of the recorded run: 18 cases executed, 18 passed, 0 failed.

Release identity: the initial release, tag sr2 - four models, 68 Camunda Forms, and the run recorded on 26 September 2026. The readable PDF exports in 02_Operational_BPMN were produced from the same snapshot.

Every scenario below is a case in that run. The scenario, its data, its expected outcome and its verdict are generated from the run record in 05_Test_Evidence/acceptance-run.json, so the plan cannot drift from the evidence.

1. Test scenarios

Case	Scenario	Process	Business rule exercised	AC ID	REQ ID	Result
TC-01	Complete referral documentation is accepted for clinical review	PR_Operational_Merged	BR-01 No clinical review until every expected document is present.	AC-01, AC-02	REQ-01, REQ-02	PASS
TC-02	Missing documentation triggers a request and a return path	PR_Operational_Merged	BR-02 A referral with a missing document is returned to the referrer and re-checked.	AC-02, AC-03	REQ-02, REQ-03	PASS
TC-03	Rejected referral closes with a notification	PR_Operational_Merged	BR-03 Only the Consultant may reject, and the referring organisation must be notified.	AC-04, AC-05	REQ-04, REQ-07	PASS
TC-04	Redirected referral is sent to another specialist service	PR_Operational_Merged	BR-04 A redirected referral goes to another specialist service, never to booking.	AC-04, AC-05	REQ-04, REQ-07	PASS
TC-05	No suitable appointment slot is escalated to the pathway coordinator	PR_Operational_Merged	BR-05 No suitable slot escalates to the Pathway Coordinator; never book outside the period.	AC-06	REQ-08, REQ-12	PASS
TC-06	Appointment within 14 days requires a telephone call	PR_Operational_Merged	BR-06 An appointment within 14 days must also be confirmed by telephone.	AC-07	REQ-09	PASS
TC-07	Appointment more than 14 days away skips the telephone call	PR_Operational_Merged	BR-07 An appointment beyond 14 days creates no telephone task.	AC-07	REQ-09	PASS
TC-08	Unsuccessful contact is recorded and a further attempt is made	PR_Operational_Merged	BR-08 Every contact attempt is recorded, and a failed attempt leads to another.	AC-08	REQ-10	PASS

Case	Scenario	Process	Business rule exercised	AC ID	REQ ID	Result
TC-09	Patient refuses consent and the pathway closes	PR_Operational_Merged	BR-09 No consent recorded, no Treatment Booking Request.	AC-09	REQ-13	PASS
TC-10	Unauthorised treatment booking request is returned to the clinical team	PR_Operational_Merged	BR-10 An unauthorised request is returned to clinical staff and never requests capacity.	AC-09	REQ-14	PASS
TC-11	Unavailable external capacity leaves the booking pending and retries	PR_Operational_Merged	BR-11 Unavailable capacity leaves the booking pending, records the attempt and creates no duplicate.	AC-06, AC-10	REQ-15, REQ-16	PASS
TC-12	Unapproved funding raises a funding approval request	PR_Operational_Merged	BR-12 No funding route determined, no confirmed appointment.	AC-10	REQ-17, REQ-18	PASS
TC-13	Happy path: referral accepted and treatment schedule confirmed	PR_Operational_Merged	BR-13 Acceptance, booking, consent, authorisation, capacity and funding all precede a confirmed schedule.	AC-01 to AC-11	REQ-06, REQ-19	PASS
TC-14	Advance payment confirmed and recorded	PR_Operational_Merged	BR-14 A required advance payment is recorded, with reference and amount, before confirmation.	AC-11	REQ-19, REQ-20	PASS
TC-15	Declined payment is handled without creating a duplicate booking	PR_Operational_Merged	BR-15 A decline notifies patient and team, allows one retry, and creates no second booking or charge.	AC-12	REQ-22	PASS
TC-16	Payment without a returned confirmation is investigated, not re-requested	PR_Operational_Merged	BR-16 A payment taken but not confirmed is investigated, never re-requested.	AC-12	REQ-23	PASS
TC-17	Urgent treatment proceeds without confirmed payment and is referred to Finance	PR_Operational_Merged	BR-17 Urgent treatment may skip payment only with the reason recorded and Finance notified.	AC-10, AC-12	REQ-24	PASS
TC-18	No advance payment required: no payment task is created	PR_Operational_Merged	BR-18 No advance payment required means no payment task and no payment	AC-10, AC-11	REQ-19	PASS

Case	Scenario	Process	Business rule exercised	AC ID	REQ ID	Result
			record.			

Each scenario is a **complete path** through the model, not a single step: the engine drives the instance from the start event to an end state, and the path taken is asserted against the expected outcome in section 5.

Identifiers. AC-nn is a criterion identifier from the acceptance criteria set for this release (AC-01 to AC-20, stated in full in the acceptance criteria evaluation). REQ-nn is the case-study requirement the criterion traces to (REQ-01 to REQ-44). BR-nn names the business rule the scenario exercises.

Pass/fail condition. Section 5 states, for every case, what the run must reach, must not reach and must end in, and what the recorded run actually produced. A case passes only if all of those hold and no incident is raised.

2. Preconditions

Common to every scenario:

1. A Camunda 8 cluster is running and reachable at http://127.0.0.1:8080/v2.

2. The four models and the 68 Camunda Forms are deployed together, one deployment per model, because every form reference carries binding="deployment" (node 09_Tests_and_Tools/deploy-all.mjs).

3. Seven process definitions are live: PR_Operational_Merged, PR_Landscape, PR_ReferralToAuthorisation, PR_TreatmentToAftercare, PR_EnquiryHandling, PR_ManagementReporting, PR_ReferringOrganisation. PR_ReferringOrganisation is the collaborating referring-organisation process inside the operational collaboration; it is started only when that pool sends a referral.

4. The external worker service is running, so the automated activities are done: 04_Java_Worker/run-workers.ps1 reports 13 workers subscribed and polling.

5. No earlier instance of the same scenario is still active under the same patientId; each case uses a unique one.

6. Fault injection is set per scenario through workers.properties or the environment (HPAS_FAILURE_MODE, HPAS_FAILURE_PREFIX), never by editing code.

Scenario-specific differences are the start variables in section 3.

3. Test data

The data is supplied in two places: the **start variables** that open the instance, and the **decisions** the human role makes at each user task. Everything else is derived by the model or written by the workers.

Case	Start variables	Human decisions at the user tasks
TC-01	`patientId=P-TC-01, documentsComplete=True`	(none)
TC-02	`patientId=P-TC-02, documentsComplete=False`	`documentsComplete=True, referralDecision=accepted, referralUrgency=routine, suitableSlotFound=True, appointmentWithin14Days=False, consentGiven=True, requestAuthorised=True, treatmentCapacityConfirmed=True, fundingApproved=True, advancePaymentRequired=False`
TC-03	`patientId=P-TC-03, documentsComplete=True`	`documentsComplete=True, referralDecision=rejected`
TC-04	`patientId=P-TC-04, documentsComplete=True`	`documentsComplete=True, referralDecision=redirected`
TC-05	`patientId=P-TC-05, documentsComplete=True`	`documentsComplete=True, referralDecision=accepted, referralUrgency=routine, suitableSlotFound=True, appointmentWithin14Days=False, consentGiven=True, requestAuthorised=True, treatmentCapacityConfirmed=True, fundingApproved=True, advancePaymentRequired=False`
TC-06	`patientId=P-TC-06, documentsComplete=True`	`documentsComplete=True, referralDecision=accepted, referralUrgency=routine, suitableSlotFound=True,`

Case	Start variables	Human decisions at the user tasks
		appointmentWithin14Days=True, contactSuccessful=True, consentGiven=True, requestAuthorised=True, treatmentCapacityConfirmed=True, fundingApproved=True, advancePaymentRequired=False
TC-07	patientId=P-TC-07, documentsComplete=True	documentsComplete=True, referralDecision=accepted, referralUrgency=routine, suitableSlotFound=True, appointmentWithin14Days=False, consentGiven=True, requestAuthorised=True, treatmentCapacityConfirmed=True, fundingApproved=True, advancePaymentRequired=False
TC-08	patientId=P-TC-08, documentsComplete=True	documentsComplete=True, referralDecision=accepted, referralUrgency=routine, suitableSlotFound=True, appointmentWithin14Days=True, contactSuccessful=True, consentGiven=True, requestAuthorised=True, treatmentCapacityConfirmed=True, fundingApproved=True, advancePaymentRequired=False
TC-09	patientId=P-TC-09, documentsComplete=True	documentsComplete=True, referralDecision=accepted, referralUrgency=routine, suitableSlotFound=True, appointmentWithin14Days=False, consentGiven=False
TC-10	patientId=P-TC-10, documentsComplete=True	documentsComplete=True, referralDecision=accepted, referralUrgency=routine, suitableSlotFound=True, appointmentWithin14Days=False, consentGiven=True, requestAuthorised=True, treatmentCapacityConfirmed=True, fundingApproved=True, advancePaymentRequired=False
TC-11	patientId=P-TC-11, documentsComplete=True	documentsComplete=True, referralDecision=accepted, referralUrgency=routine, suitableSlotFound=True, appointmentWithin14Days=False, consentGiven=True, requestAuthorised=True, treatmentCapacityConfirmed=True, fundingApproved=True, advancePaymentRequired=False
TC-12	patientId=P-TC-12, documentsComplete=True	documentsComplete=True, referralDecision=accepted, referralUrgency=routine, suitableSlotFound=True, appointmentWithin14Days=False, consentGiven=True, requestAuthorised=True, treatmentCapacityConfirmed=True, fundingApproved=True, advancePaymentRequired=False
TC-13	patientId=P-TC-13, documentsComplete=True	documentsComplete=True, referralDecision=accepted, referralUrgency=routine, suitableSlotFound=True, appointmentWithin14Days=False, consentGiven=True, requestAuthorised=True, treatmentCapacityConfirmed=True, fundingApproved=True, advancePaymentRequired=False

Case	Start variables	Human decisions at the user tasks
TC-14	patientId=P-TC-14, documentsComplete=True	documentsComplete=True, referralDecision=accepted, referralUrgency=routine, suitableSlotFound=True, appointmentWithin14Days=False, consentGiven=True, requestAuthorised=True, treatmentCapacityConfirmed=True, fundingApproved=True, advancePaymentRequired=True, paymentStatus=confirmed
TC-15	patientId=P-TC-15, documentsComplete=True	documentsComplete=True, referralDecision=accepted, referralUrgency=routine, suitableSlotFound=True, appointmentWithin14Days=False, consentGiven=True, requestAuthorised=True, treatmentCapacityConfirmed=True, fundingApproved=True, advancePaymentRequired=True, paymentStatus=confirmed
TC-16	patientId=P-TC-16, documentsComplete=True	documentsComplete=True, referralDecision=accepted, referralUrgency=routine, suitableSlotFound=True, appointmentWithin14Days=False, consentGiven=True, requestAuthorised=True, treatmentCapacityConfirmed=True, fundingApproved=True, advancePaymentRequired=True, paymentStatus=no response
TC-17	patientId=P-TC-17, documentsComplete=True	documentsComplete=True, referralDecision=accepted, referralUrgency=routine, suitableSlotFound=True, appointmentWithin14Days=False, consentGiven=True, requestAuthorised=True, treatmentCapacityConfirmed=True, fundingApproved=True, advancePaymentRequired=True, paymentStatus=urgent clinical need
TC-18	patientId=P-TC-18, documentsComplete=True	documentsComplete=True, referralDecision=accepted, referralUrgency=routine, suitableSlotFound=True, appointmentWithin14Days=False, consentGiven=True, requestAuthorised=True, treatmentCapacityConfirmed=True, fundingApproved=True, advancePaymentRequired=False

Which gateway each decision drives

Decision variable	Gateway in the model
documentsComplete	G_DocumentsComplete - "All expected supporting documents available?"
referralDecision	G_ReferralDecision - "Referral decision?"
referralUrgency	G_UrgentReferral - "Urgent referral requiring expedited booking?"
suitableSlotFound	G_SlotsAvailable - "Suitable appointment available within the requested period?"
appointmentWithin14Days	G_WithinFourteenDays - "Appointment due within the next 14 days?"
contactSuccessful	G_ContactSuccessful - "Patient confirmed the appointment?"
consentGiven	G_PatientConsent - "Patient consents to proceed with treatment?"
requestAuthorised	G_RequestAuthorised - "Request completed and authorised by an authorised clinical professional?"
treatmentCapacityConfirmed	G_CapacityConfirmed - "External treatment capacity confirmed?"
fundingApproved	G_FundingApproved - "Funding approved or no payment required?"
advancePaymentRequired	G_AdvancePaymentRequired - "Advance payment required before the

Decision variable	Gateway in the model
	appointment is confirmed?"
`paymentStatus`	G_PaymentOutcome - "Payment outcome returned by the payment service?"

4. Actions

For each scenario the same sequence is performed, by `09_Tests_and_Tools/run-acceptance.mjs`:

1. Publish the start message (or create the instance directly where the process has a plain start event) with the case's start variables.

2. Wait for the instance to appear and read its state.

3. Whenever a user task is created, complete it through the API with the decisions listed for that case, in the order the instance asks for them.

4. Let the worker service finish every automated activity; if a scenario exercises a failure path, set the fault for that activity first.

5. Repeat steps 3 and 4 until the instance reaches an end state or the case times out.

6. Record the elements reached with their occurrence counts, the end state, the variables, and any incident.

7. Compare the record with the expected outcome and mark the case PASS or FAIL.

5. Expected outcomes

TC-01 — Complete referral documentation is accepted for clinical review

Process: `PR_Operational_Merged`

- must reach: `T_ClinicalReviewReferral`, `T_RecordReferralDecision`
- actual: end state **COMPLETED**, 0 incident(s) — **PASS**

TC-02 — Missing documentation triggers a request and a return path

Process: `PR_Operational_Merged`

- must reach: `T_RequestMissingDocuments`, `T_ClinicalReviewReferral`
- must reach `T_CheckReferralDocuments` at least **2 times** (the loop is the point of the case)
- actual: end state **COMPLETED**, 0 incident(s) — **PASS**

TC-03 — Rejected referral closes with a notification

Process: `PR_Operational_Merged`

- must reach: `T_NotifyReferralOutcome`
- must **not** reach: `T_ForwardAcceptedReferral`
- must end in state **COMPLETED**
- actual: end state **COMPLETED**, 0 incident(s) — **PASS**

TC-04 — Redirected referral is sent to another specialist service

Process: `PR_Operational_Merged`

- must reach: `T_RedirectReferral`
- must end in state **COMPLETED**
- actual: end state **COMPLETED**, 0 incident(s) — **PASS**

TC-05 — No suitable appointment slot is escalated to the pathway coordinator

Process: `PR_Operational_Merged`

- must reach: `T_ResolveCapacity`, `T_SelectAndBookSlot`
- must reach `T_QueryAvailableSlots` at least **2 times** (the loop is the point of the case)
- actual: end state **COMPLETED**, 0 incident(s) — **PASS**

TC-06 — Appointment within 14 days requires a telephone call

Process: `PR_Operational_Merged`

- must reach: `T_TelephonePatient`, `T_RecordContactAttempt`
- actual: end state **COMPLETED**, 0 incident(s) — **PASS**

TC-07 — Appointment more than 14 days away skips the telephone call

Process: `PR_Operational_Merged`

- must reach: T_NewPatientConsultation
- must **not** reach: T_TelephonePatient
- actual: end state **COMPLETED**, 0 incident(s) — **PASS**

TC-08 — Unsuccessful contact is recorded and a further attempt is made

Process: PR_Operational_Merged

- must reach: T_TelephonePatient
- must reach T_TelephonePatient at least **2 times** (the loop is the point of the case)
- actual: end state **COMPLETED**, 0 incident(s) — **PASS**

TC-09 — Patient refuses consent and the pathway closes

Process: PR_Operational_Merged

- must reach: T_AdviseNoTreatment
- must **not** reach: T_CreateTreatmentBookingRequest
- must end in state **COMPLETED**
- actual: end state **COMPLETED**, 0 incident(s) — **PASS**

TC-10 — Unauthorised treatment booking request is returned to the clinical team

Process: PR_Operational_Merged

- must reach: T_ReturnUnauthorisedRequest
- must reach T_CreateTreatmentBookingRequest at least **2 times** (the loop is the point of the case)
- actual: end state **COMPLETED**, 0 incident(s) — **PASS**

TC-11 — Unavailable external capacity leaves the booking pending and retries

Process: PR_Operational_Merged

- must reach: T_KeepPending
- must reach T_RequestTreatmentCapacity at least **2 times** (the loop is the point of the case)
- actual: end state **COMPLETED**, 0 incident(s) — **PASS**

TC-12 — Unapproved funding raises a funding approval request

Process: PR_Operational_Merged

- must reach: T_RequestFundingApproval, T_RecordFundingApproval
- must reach T_DetermineFundingRoute at least **2 times** (the loop is the point of the case)
- actual: end state **COMPLETED**, 0 incident(s) — **PASS**

TC-13 — Happy path: referral accepted and treatment schedule confirmed

Process: PR_Operational_Merged

- must reach: T_ConfirmTreatmentSchedule
- must end in state **COMPLETED**
- actual: end state **COMPLETED**, 0 incident(s) — **PASS**

TC-14 — Advance payment confirmed and recorded

Process: PR_Operational_Merged

- must reach: T_SendSecurePaymentRequest, T_RecordPayment
- must end in state **COMPLETED**
- actual: end state **COMPLETED**, 0 incident(s) — **PASS**

TC-15 — Declined payment is handled without creating a duplicate booking

Process: PR_Operational_Merged

- must reach: T_HandlePaymentFailure, T_RecordPayment
- must reach T_SendSecurePaymentRequest at least **2 times** (the loop is the point of the case)
- actual: end state **COMPLETED**, 0 incident(s) — **PASS**

TC-16 — Payment without a returned confirmation is investigated, not re-requested

Process: PR_Operational_Merged

- must reach: T_MarkForInvestigation, T_ResolveTransaction
- must end in state **COMPLETED**
- actual: end state **COMPLETED**, 0 incident(s) — **PASS**

TC-17 — Urgent treatment proceeds without confirmed payment and is referred to Finance

Process: PR_Operational_Merged

- must reach: `T_AuthoriseUrgentTreatment`, `T_ReferUnconfirmedPayment`
- must end in state **COMPLETED**
- actual: end state **COMPLETED**, 0 incident(s) — **PASS**

TC-18 — No advance payment required: no payment task is created

Process: `PR_Operational_Merged`

- must reach: `T_ConfirmTreatmentSchedule`
- must **not** reach: `T_SendSecurePaymentRequest`
- must end in state **COMPLETED**
- actual: end state **COMPLETED**, 0 incident(s) — **PASS**

6. Pass / Fail conditions

A case passes only when **all** of the following hold:

1. **Every** element in `must reach` appears in the instance history. 2. **No** element in `must not reach` appears — this is what makes the negative branches testable, not just the happy path. 3. Every `minOccurrences` count is met, which is how a loop is proved to have looped. 4. The end state equals the expected end state. 5. **Zero incidents** were raised. An unhandled-error incident means a worker rejected a job or a variable was missing, so the path taken is not evidence of anything. 6. The run completes within the case timeout.

A case fails on the first of these that does not hold, and the failing condition is named in `problems`.

Recorded result: 18 of 18 cases passed, 0 failed.

7. Environment and how to reproduce

Item	Value
Execution	`node 09_Tests_and_Tools/run-acceptance.mjs`
Record	`05_Test_Evidence/acceptance-run.json`
Log	`05_Test_Evidence/acceptance-run.log`
Models executed	PR_EnquiryHandling, PR_Landscape, PR_ManagementReporting, PR_Operational_Merged, PR_ReferralToAuthorisation, PR_TreatmentToAftercare
One-command check of everything	`09_Tests_and_Tools/verify-all.ps1` (9 steps, all pass)